

Classification Metrics: Three Banking Examples

A simple guide to confusion matrix, precision, recall, F1 score, ROC curve, and AUC. All figures below are teaching examples, not measured bank results.

1. First, choose the positive class

Positive means the event we want to find. For fraud it is a fraudulent transaction; for loan risk it is a default; for an offer it is a customer who accepts. The meaning of each metric depends on this choice.

2. Confusion matrix

A confusion matrix compares what actually happened with what a model predicted. It counts correct and incorrect predictions in four groups:

Actual / prediction	Predicted positive	Predicted negative
Actually positive	TP: correctly found positive	FN: missed positive
Actually negative	FP: false alarm	TN: correctly found negative

Read across the actual row first. TP and TN are correct; FP and FN are errors. In banking, which error hurts more depends on the decision.

3. What each score answers

Measure	Plain-language question	Formula
Precision	When the model says positive, how often is it right?	$TP / (TP + FP)$
Recall (sensitivity)	Of all real positives, how many did it find?	$TP / (TP + FN)$
F1 score	How well does it balance precision and recall?	$2 \times \text{precision} \times \text{recall} / (\text{precision} + \text{recall})$
False positive rate	Of real negatives, how many were falsely flagged?	$FP / (FP + TN)$

An F1 score is high only when precision and recall are both reasonably high. A score of 0 is worst; 1 is best. Accuracy, used later for comparison, is $(TP + TN) / \text{all cases}$.

4. ROC curve and AUC

Many models produce a probability, such as 0.80 chance of fraud. A cutoff converts that probability into a yes/no prediction. Lowering the cutoff usually catches more real positives, but also raises more false alarms. The ROC curve plots recall (vertical axis) against false positive rate (horizontal axis) for many cutoffs.

AUC is the area under that ROC curve: 0.5 is roughly random ranking and 1.0 is perfect ranking. AUC summarizes how well the model ranks positives above negatives across cutoffs. It does not choose a cutoff or tell us the exact precision at the cutoff we will use. ROC-AUC requires probabilities or scores, not just yes/no predictions.

Example 1: Fraud detection

Positive = fraudulent transaction. A bank examines 1,000 transactions: 100 fraud and 900 genuine. At its chosen cutoff, the model flags 120 transactions.

Step 1 - Count the outcomes

Actual / prediction	Predicted positive	Predicted negative
Actually positive	TP = 80	FN = 20
Actually negative	FP = 40	TN = 860

Step 2 - Calculate and interpret

Metric	Calculation	Meaning here
Precision	$80 / (80 + 40) = 66.7\%$	Of predicted positives, this percentage is truly positive.
Recall	$80 / (80 + 20) = 80.0\%$	Of real positives, this percentage was found.
F1 score	72.7%	Combined precision and recall at this cutoff.
False positive rate	$40 / 900 = 4.4\%$	Share of real negatives incorrectly flagged.

Step 3 - Make a banking decision

The team catches 80 of 100 frauds but reviews 40 genuine transactions. If missing fraud is very costly, the bank may accept more false alarms.

Step 4 - Understand ROC-AUC

AUC cannot be calculated from this one confusion matrix. The bank needs fraud scores for all 1,000 cases and must test multiple cutoffs.

Example 2: Loan default prediction

Positive = borrower defaults. A bank reviews 200 past loans: 50 defaulted and 150 did not. The model labels 60 borrowers high risk.

Step 1 - Count the outcomes

Actual / prediction	Predicted positive	Predicted negative
Actually positive	TP = 35	FN = 15
Actually negative	FP = 25	TN = 125

Step 2 - Calculate and interpret

Metric	Calculation	Meaning here
Precision	$35 / (35 + 25) = 58.3\%$	Of predicted positives, this percentage is truly positive.
Recall	$35 / (35 + 15) = 70.0\%$	Of real positives, this percentage was found.
F1 score	63.6%	Combined precision and recall at this cutoff.
False positive rate	$25 / 150 = 16.7\%$	Share of real negatives incorrectly flagged.

Step 3 - Make a banking decision

The model detects 35 of 50 defaults. It also flags 25 borrowers who would have repaid; an automatic rejection could unfairly affect them. The bank should choose its cutoff carefully and review decisions.

Step 4 - Understand ROC-AUC

To draw ROC, score every borrower, try several cutoffs, and plot recall against false positive rate. The single operating point here is (16.7%, 70.0%).

Example 3: Term deposit offer

Positive = customer subscribes. A bank contacts 500 customers: 100 subscribe and 400 do not. The model predicts 100 subscribers.

Step 1 - Count the outcomes

Actual / prediction	Predicted positive	Predicted negative
Actually positive	TP = 60	FN = 40
Actually negative	FP = 40	TN = 360

Step 2 - Calculate and interpret

Metric	Calculation	Meaning here
Precision	$60 / (60 + 40) = 60.0\%$	Of predicted positives, this percentage is truly positive.
Recall	$60 / (60 + 40) = 60.0\%$	Of real positives, this percentage was found.
F1 score	60.0%	Combined precision and recall at this cutoff.
False positive rate	$40 / 400 = 10.0\%$	Share of real negatives incorrectly flagged.

Step 3 - Make a banking decision

Of 100 customers predicted to subscribe, 60 do. The model misses 40 subscribers. A marketing team may care about precision when calls are expensive, or recall when reaching more potential subscribers matters.

Step 4 - Understand ROC-AUC

To compare targeting cutoffs, rank customers by subscription score and compute an ROC curve. The point shown here is (10.0%, 60.0%).

Compare the three examples

Use case	Precision	Recall	F1	What matters in practice
Fraud	66.7%	80.0%	72.7%	Catch fraud; manage investigation workload.
Loan default	58.3%	70.0%	63.6%	Catch risk; avoid harming reliable borrowers.
Term deposit	60.0%	60.0%	60.0%	Balance reach against cost of contact.

These percentages belong to different problems and populations, so the highest number does not automatically mean the best business model. Choose the positive class and the cost of each mistake first.

Quick memory aid

Precision: When we say yes, are we right? Recall: Of the real yes cases, how many did we find? F1: Are precision and recall balanced? ROC: What happens to detection and false alarms as we move the cutoff? AUC: How well are positives ranked above negatives overall?

Optional accuracy check

Fraud: $(80 + 860) / 1,000 = 94\%$. Yet 20 frauds are missed. This is why accuracy alone can hide important errors when most transactions are genuine.